Difference between two independent sample means

1 RECOGNISE IT

- Two independent random samples are compared.
- Event concerns $\bar{X}_1-\bar{X}_2$.
- Population sigmas are known; populations normal or both n values are large.

2 DECISION / METHOD

1. Fix order: group 1 minus group 2.
2. Centre at $\mu_1-\mu_2$.
3. Combine variances, not SDs.
4. Standardise the difference and use normal areas.

3 FORMULA

$E(\bar{X}_1-\bar{X}_2)=\mu_1-\mu_2$

$SE=\sqrt{\sigma_1^2/n1+\sigma_2^2/n2}$

$z=[(d)-(\mu_1-\mu_2)]/SE$

4 FORMULA SHEET VS WHAT I KNOW

Gives: statistic, parameter difference, combined SE and Z when both sigmas are known.
Still know: independence, fixed subtraction order, and event direction.

5 EXCEL ROUTE

For $P(\bar{X}_1-\bar{X}_2>c)$:

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=1-NORM.DIST(c,mu1-mu2,SQRT(sigma1^2/n1+sigma2^2/n2),TRUE)
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6 TINY WORKED EXAMPLE

$\mu_1-\mu_2=5$, $\sigma_1=3$,$n1=25$,$\sigma_2=4$,$n2=16$. $SE=\sqrt{1.36}=1.166$. $P(diff>7)$ uses $z=1.715$, about .0432.

7 TOP EXAM TRAPS

- Reversing group order mid-solution.
- Adding SDs instead of variances.
- Using paired methods for independent samples.

8 SAY ALOUD

For independent means, subtract in one fixed order and add their sampling variances under the square root.

9 EXIT CHECK

What is the centre of $\bar{X}_1-\bar{X}_2$?

ONE-GLANCE VISUAL ROUTE

LOCK: group 1-group 2 → **add variances** → **divide by SE** → **Normal area**